W_β *Affects* $Net_j \Rightarrow$

Net_j *Affects* $O_j \Rightarrow$

O_j *Affects* $(Net_{k=1}, Net_{k=2}, ..., Net_{k=m}) \Rightarrow$

$(Net_{k=1}, Net_{k=2}, ..., Net_{k=m})$ *Affect Error*



$$\left[\frac{\partial Net_{k=1}}{\partial O_j}\right] + \left[\frac{\partial E}{\partial Net_{k=2}} \cdot \frac{\partial Net_{k=2}}{\partial O_j}\right] + \ldots + \left[\frac{\partial E}{\partial Net_{k=m}} \cdot \frac{\partial Net_{k=m}}{\partial O_j}\right] \cdot \left[\frac{\partial O_j}{\partial Net_j}\right] \cdot \left[\frac{\partial}{}\right.$$

$$\frac{\partial E}{\partial Net_k} \rightarrow -\delta_k$$

$$\text{Net}_k = \sum_k w_{kj} O_j \Rightarrow \frac{\partial \text{Net}_k}{\partial O_j} = w_{kj}$$

$$O_j = \frac{1}{1+e^{-Net_j}} \rightarrow \frac{\partial O_j}{\partial Net_j} = O_j(1-O_j$$

$$\text{Net}_j = \sum_j w_{ij} O_i \Rightarrow \frac{\partial \text{Net}_j}{\partial w_{ji}} = O_i$$